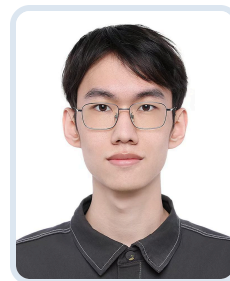


Xinyu Chen

Beijing, China chenxinyu20040529@gmail.com +86 18911827529



Education

**Peking University, General Artificial Intelligence Experimental Class,
Yuanpei College**

Sept 2022 - Jul 2026

- **Program:** General Artificial Intelligence Experimental Class, mentored by Prof. Song-Chun Zhu, a renowned AI scholar in China and abroad.
- **Major:** Artificial Intelligence
- **GPA:** 3.565/4.0 | 85/100 (first three years: 3.667/4.0 | 86.7/100)
- **Core Courses:** Machine Learning (91), Mathematical Foundation for AI (90), Multi-agent System (94), Introduction to Computer Vision (93), AI Systems Practice I (96), AI Systems Practice II (97), Cognition and Reasoning (90.5), Deep Learning and Natural Language Processing (93)

Beijing 101 High School

Sept 2016 - Jul 2022

- Ranked in the top 20 in Beijing in the National College Entrance Examination.

**Peking University, School of Computer
Science, Ph.D. Student**

Planned enrollment Sept 2026; graduation Jul 2031

- **Advisor:** Tengjiao Wang, Chang Jiang Scholar, Director of the PKU Center for Computational Social Science, Dean of the PKU (Qingdao) Institute of Computational Social Science, and Ph.D. supervisor.

PAIR Lab | Lab Member

Peking University Alignment and Interaction Research Lab, focusing on reinforcement learning, multi-agent systems, game theory, and AGI value alignment.

BAAI | Intern

A leading nonprofit AI research institute in China, focusing on foundation models, embodied intelligence, AI safety, and open-source infrastructure.

English Proficiency

CET-4: 605 (Dec 2022)

CET-6: 664 (Jun 2023)

TOEFL: 107 (Oct 2023)

Languages & Skills

- **Programming Languages:** C++, C, CUDA, Python
- **Natural Languages:** Chinese (Native), English (Proficient)
- Exchanged at the National University of Singapore for one semester from Jan 2025 to May 2025 and adapted to fully English-taught coursework.
- Attended NeurIPS 2025 in San Diego, USA in Dec 2025 and developed English academic communication capability.

Selected Awards

- **2024:** Xingzhengde Scholarship (¥5000 RMB)
- **2023:** Peking University Social Work Award
- **2022:** Linhu Scholarship (¥50000 RMB)

Research Experience

AI for Law, Peking University / BIGAI

Mar 2024 - Jun 2024

- Constructed a dataset for structured extraction of key legal case elements.
- Demonstrated that fine-tuned large language models can extract critical features of legal cases to support model-based case judgments.

Safety Evaluation of Large Language Models, PAIR Lab

Sept 2024 - Nov 2024

- Explored the DNA paradigm for category-agnostic and scalable all-round safety evaluation of large language models.

LLM & VLM Alignment, PAIR Lab

Jun 2024 - May 2025

- Contributed to the *Align Anything* framework, including training code, interactive terminals, and datasets.
- Developed the Beaver-V preference dataset to improve the safety and helpfulness of vision-language models.
- Designed the Safe RLHF-V algorithm to balance VLM safety and helpfulness.

LLM-based Social Relation Understanding, Peking University / BIGAI

Dec 2024 - May 2025

- Collaborated with Fanqi Kong to explore adversarial model generation methods for improving bot detection in large language models.

LLM Math Agent

Sept 2025 - Present

- Working on automated mathematical proof verifiers based on step decomposition and fine-grained review.

Multi Agent LLM

Sept 2025 - Present

- Working on multi-agent reinforcement learning methods to improve long-context capabilities of language models.

LLM LoRA Memory

May 2026 - Present

- Developing new methods to substantially improve model capabilities during test-time training.

AI for Perovskite

Sept 2025 - Present

- Using fine-tuned pretrained molecular models to predict delta PCE for dopant molecules in perovskite solar cells.

AI for Cryo-ET Reconstruction

Sept 2025 - Present

- Building a variational-inference-based Cryo-ET reconstruction paradigm and demonstrating its usefulness and inspiration for the community from both theoretical and experimental perspectives.
- Developed a series of works including CryoGEN, CryoGEN-II, and CryoWGEN; completed the [Cryo-ET blog](#) and the knowledge base [Open Cryo-ET](#).

Projects

Align-Anything

[repo](#)

Project Function An open-source all-modality foundation-model alignment framework from the PKU-Alignment team, designed for post-training and alignment of LLMs, VLMs, and any-to-any multimodal models, with support for text, image, audio, and video input/output modalities.

Personal Contribution Implemented multiple LLM alignment algorithms including ORPO, SimPO, KTO, and PEFT; led development of the interactive terminal; contributed to the [align-anything dataset](#).

Safe-RLHF-V / BeaverTails-V

[repo](#)

Project Function Extends traditional Safe RLHF to vision-language scenarios by separately modeling helpfulness reward and safety cost, then applying Lagrangian constrained optimization to improve usefulness while controlling safety risks. It also introduces the BeaverTails-V multimodal safety preference dataset and the Beaver-Guard-V multi-level safety guard system for MLLM safety training, filtering, and evaluation.

Personal Contribution As a major contributor, implemented Safe-RLHF-V training and evaluation code and built the multimodal preference dataset BeaverTails-V.

AgentMind

Agent platform

Project Function A full-process agent platform for data collection, knowledge-base RAG graph construction, and deep research. It supports multi-source data collection, RAG and entity-enhanced graph construction, and deep research workflows across planning, collection, analysis, verification, and report generation.

Personal Contribution As a major contributor, led development of the deep research module, RAG inference acceleration module, and related LLM tools and skills.

Undergraduate Thesis: Implementation of LLM-based Multimodal Semantic Compression

Undergraduate Thesis

Project Function Focused on multimodal semantic compression in large-language-model scenarios; designed and implemented an intelligent compression and retrieval-augmented system for long documents and interleaved text-image data. Through task-aware schema generation, context-aware text-image fusion, and traceable hierarchical indexing, the system improves knowledge-base semantic density, retrieval efficiency, and generation reliability.

Personal Contribution Completed and defended the thesis, receiving an **Excellent** evaluation.

Contest

Innovation Category at the 9th National Youth AI Innovation and Entrepreneurship Conference

Dec 2024

Team Leader of Zhichi, First Prize | 5-person team

Project Function Proposed an ensemble-learning model to predict the spatial relationship between the third molar and mandibular canal, surpassing professional dental experts; the system assists clinical dentists in preoperative diagnosis for wisdom tooth surgery.

Personal Contribution As team leader, traveled to Shenzhen to present the project and successfully completed the project defense.

Research Outputs

Safe RLHF-V: Safe Reinforcement Learning from Multi-modal Human Feedback

NeurIPS 2025

Authors: Jiaming Ji, Xinyu Chen, Rui Pan, Conghui Zhang, Han Zhu, Jiahao Li, Donghai Hong, Boyuan Chen, Jiayi Zhou, Kaile Wang, Juntao Dai, Chi-Min Chan, Yida Tang, Sirui Han, Yike Guo, Yaodong Yang.

Summary: Safe RLHF-V studies safety alignment for multimodal large language models, addressing harmful responses in image-text scenarios and the limitation that traditional single-preference RLHF struggles to balance helpfulness and safety. The paper introduces BeaverTails-V, Beaver-Guard-V, and a Safe RLHF-V algorithm based on decoupled reward/cost modeling and Lagrangian constrained optimization.

Personal Contribution: As co-first author, responsible for most code implementation, dataset construction, and evaluation.

CryoGEN-II: Cryogenic Electron Tomography Reconstruction via Generative Network

CVPR 2026 Workshop

Authors: Yunfei Teng, Xinyu Chen, He Zhang, Qiwei Ye, Yulei Hung, Sha Hu.

Summary: CryoGEN-II is a generative method for Cryo-ET 3D reconstruction, aiming to mitigate the missing-wedge problem caused by limited-angle acquisition and the resulting Z-axis elongation, artifacts, and structural distortion. The paper reinterprets Cryo-ET reconstruction from a variational-inference perspective and introduces latent-space Sharpness-Aware Minimization to improve training stability and reconstruction robustness.

Personal Contribution: As co-first author, responsible for in-situ Cryo-ET algorithm implementation and experiments.

CryoWGEN: Cryogenic Electron Tomography Reconstruction via Wasserstein Generative Network

In submission

Authors: Xinyu Chen, Zhichao Chen, Hongwang Xiao, Xi Chen, Yuxuan Xiong, Yan Wang, Zhouchen Lin, Yunfei Teng.

Summary: CryoWGEN proposes a new generative reconstruction framework that reformulates missing-wedge completion as variational inference and further introduces optimal transport, enabling the model to learn posterior distributions that recover isotropic resolution while preserving structural fidelity and interpretability.

Personal Contribution: As first author, responsible for all algorithm implementation, in-situ Cryo-ET experiments, and part of the paper writing.

EVIA: Entropic Variational Inference Auto-encoding

UAI 2026

Authors: Yunfei Teng, Zhichao Chen, Xinyu Chen, Lulu Tang, Sixin Zhang, Zhouchen Lin.

Summary: EVIA is an auto-encoding generative framework combining entropy regularization, optimal transport, and variational inference. It extends the deterministic latent mapping in WAE to distribution-level posterior modeling and uses SGLD sampling to improve training stability and latent-space alignment, outperforming VAE/AAE/WAE on posterior estimation, image generation, and image inpainting.

Personal Contribution: Responsible for improving experimental results and writing the experimental section.

Enhancing LLM-Based Social Bot via an Adversarial Learning Framework

EMNLP 2025 Oral

Authors: Fanqi Kong, Xiaoyuan Zhang, Xinyu Chen, Yaodong Yang, Song-Chun Zhu, Xue Feng.

Summary: EvoBot is an LLM-based social media agent framework designed to generate human-like content grounded in real user profiles and social relationships. The paper proposes an adversarial learning paradigm that first uses real social media data for supervised fine-tuning, then introduces a co-evolving social bot detector as dynamic feedback and optimizes EvoBot through DPO.

Personal Contribution: Responsible for part of the experiments, paper writing, and figure optimization.

SIV-Bench: A Video Benchmark for Social Interaction Understanding and Reasoning

ACL 2026

Authors: Fanqi Kong, Weiqin Zu, Xinyu Chen, Yaodong Yang, Song-Chun Zhu, Xue Feng.

Summary: SIV-Bench is a video benchmark for evaluating multimodal large language models on social interaction understanding and reasoning. Built on social relationship theory, it evaluates social scene understanding, social state reasoning, and social dynamics prediction across tasks such as action recognition, expression recognition, relationship reasoning, intention reasoning, factual prediction, and counterfactual prediction.

Personal Contribution: Participated in data collection and evaluation.

Reinforced Self-play for Long-Context Understanding

In preparation

Authors: Xinyu Chen, Yiding Wang, Shijia Kang, Muhan Zhang.

Summary: Long-context Zero Reasoner is a self-play reinforcement learning framework for improving long-context language models. Without human-labeled data, it automatically generates and solves Needle-in-a-Haystack long-context tasks to build a self-verifiable training loop, improving information localization and reasoning at million-token context lengths and alleviating the lost-in-the-middle problem.

Personal Contribution: Responsible for experimental code implementation, evaluation, and paper writing.

GraphVerify: A Dependency-Aware Multi-Agent Verifier for Natural Language Proofs

In preparation

Authors: Haotong Yang, Xinyu Chen, Huijun Yu, Wenkai Yu, Songshen He, Siqi Yang, Zhouchen Lin, Muhan Zhang.

Summary: GraphVerify is a multi-agent framework for verifying mathematical natural-language proofs. It parses proofs into Proof Dependency Graphs and dispatches specialized agents to verify each reasoning step, while integrating external tools to improve reliability and produce interpretable diagnostic reports.

Personal Contribution: Responsible for building the project code, experiments, and paper writing.

Transfer learning discovery of molecular modulators for perovskite solar cells

In preparation

Authors: Haoming Yan, Xinyu Chen, Yanran Wang, Zhangyuchang Lu, Zhengchao Luo, Weizheng Huang, Tianyu Huang, Hongshuai Wang, Peng Chen, Ziming Chen, Yuzhi Zhang, Weijie Sun, Jinzhuo Wang, Rui Zhu, Lichen Zhao.

Summary: This work uses transfer learning to discover molecular modulators for perovskite solar cells. It builds a chemistry-guided transfer learning framework based on pretrained deep neural networks to predict the impact of molecular modulators on PCE and Δ PCE, compares molecular representations, screens 79,043 commercially available molecules, and validates chemical mechanisms through interpretability analysis and experiments.

Personal Contribution: Responsible for project code implementation and part of the experiments and paper writing.

Talks

CryoGEN-II: Cryogenic Electron Tomography Reconstruction via Generative Network

Introduced the history, core workflow, and traditional methods of cryo-electron microscopy, then presented the CryoGEN/CryoGEN-II series and the new perspective of understanding Cryo-ET reconstruction as variational inference.

Blogs

Cryo-ET blog

A survey of Cryo-ET reconstruction methods, covering the cryo-electron tomography workflow, classical WBP/SIRT algorithms, and deep learning methods such as IsoNet, DeepDeWedge, and CryoGEN, with emphasis on missing-wedge correction, denoising, and 3D tomogram reconstruction.